Methodology Note: Case-Shiller Index for São Paulo Residential Real Estate

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1. Total Return Decomposition

Let P_t denote the transaction price of a property at time t, y_{t+k} the net rental income at time $t+k$, c_{t+k} the maintenance and renovation expenditures at time $t+k$, τ_{t+k} the tax payments (property tax, transfer tax, capital gains tax) at time $t+k$, and i a reinvestment rate. The total return from holding a property between purchase date t and sale date T is:

$$R_{t\to T} = \underbrace{\frac{P_T - P_t}{P_t}}_{\text{(i) Capital gains}} + \underbrace{\frac{\sum_{k=0}^{T-t} y_{t+k}(1+i)^{T-t-k}}{P_t}}_{\text{(ii) Rental yields}} - \underbrace{\frac{\sum_{k=0}^{T-t} c_{t+k}(1+i)^{T-t-k}}{P_t}}_{\text{(iii) Maintenance \& renovation}} - \underbrace{\frac{\sum_{k=0}^{T-t} \tau_{t+k}(1+i)^{T-t-k}}{P_t}}_{\text{(iv) Taxes}} . \tag{1}$$

The index constructed in this study captures only component (i). Components (ii)–(iv) are not observed in our data.

2. Repeat-Sales Index Construction

We follow the repeat-sales methodology introduced by Bailey, Muth, and Nourse (1963) and refined by Case and Shiller (1987, 1989). For a comprehensive review and comparison of repeat-sales variants, see Nagaraja, Brown, and Wachter (2014). The procedure uses properties observed in at least two transactions. For property n purchased at time t_n and sold at time s_n, the log return is:

$$r_n = \ln P_{s_n} - \ln P_{t_n} \ . \tag{2}$$

Assuming a latent log price index $\{\beta_j\}_{j=1}^{T}$ with $\beta_0 = 0$, the repeat-sales model is:

$$r_n = \beta_{s_n} - \beta_{t_n} + \varepsilon_n, \qquad n = 1, \ldots, N \ . \tag{3}$$

In matrix form, $\mathbf{r} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, where $\mathbf{X}$ is the $N \times (T-1)$ design matrix with $X_{n,j} = \mathbf{1}[j = s_n] - \mathbf{1}[j = t_n]$ for each month $j \geq 1$, and the first column is dropped for identification.

Stage 1 (OLS/BMN). The initial estimate is:

$$\hat{\boldsymbol{\beta}}^{\mathrm{OLS}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{r} \ . \tag{4}$$

Stage 2 (Heteroskedasticity estimation). Following Case and Shiller (1989), the variance of the error term is modeled as a function of the holding period. Let h_n be the holding period (in months) for pair n. The squared OLS residuals $\hat{e}_n^2 = (r_n - \mathbf{x}_n'\hat{\boldsymbol{\beta}}^{\mathrm{OLS}})^2$ are regressed on a constant and h_n:

$$\hat{e}_n^2 = \alpha_0 + \alpha_1 h_n + \eta_n \ . \tag{5}$$

The fitted values $\hat{\sigma}_n^2 = \hat{\alpha}_0 + \hat{\alpha}_1 h_n$ provide the estimated variances. The GLS weight for observation n is $w_n = 1/\hat{\sigma}_n$.

Stage 3 (GLS/Case-Shiller). The final estimate is:

$$\hat{\boldsymbol{\beta}}^{\mathrm{CS}} = (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}\mathbf{r} \ , \tag{6}$$

where $\mathbf{W} = \mathrm{diag}(w_1^2, \ldots, w_N^2)$. The index in levels is $I_j = \exp(\hat{\beta}_j^{\mathrm{CS}})$ with $I_0 = 1$.

Smoothing. A 12-month centered moving average is applied to the estimated index levels before normalization:

$$\tilde{I}_j = \frac{1}{12}\sum_{k=0}^{11} I_{j-k} \ . \tag{7}$$

The final index is normalized so that $\tilde{I}_{j^*} = 1$ at the base date j^* = January 2010.

3. Annualized Property-Level Returns

For each repeat-sale pair with purchase price P_t, sale price P_T, and holding period $\Delta = T - t$ (in years), the annualized return is:

$$r_n^{\mathrm{ann}} = \left(\frac{P_T}{P_t}\right)^{1/\Delta} - 1 \ . \tag{8}$$

The distribution reported in the main text restricts the sample to purchases between 2015 and 2019, holding periods of at least two years, and annualized returns in $[-10\%, 50\%]$.

4. Data

The data come from ITBI (*Imposto sobre Transmissão de Bens Imóveis*) records published by the São Paulo municipal government, covering approximately 2.6 million transactions (2006–2025)

involving roughly 1.5 million unique properties. After identifying repeat sales and filtering for holding periods ≥ 3 years, approximately 309,000 pairs from 263,000 distinct properties remain.

References

Bailey, M. J., Muth, R. F., and Nourse, H. O. (1963). A regression method for real estate price index construction. *Journal of the American Statistical Association*, 58(304):933–942.

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